

AGRICULTURE GUIDELINES FOR FARMERS

A Practical Guide to Scientific, Sustainable and Safe Crop Production

Introduction

Agriculture is a combination of soil management, crop selection, water management, nutrient management, pest and disease management, weather awareness, harvesting and post-harvest handling. Successful farming depends not only on selecting a suitable crop but also on understanding the conditions of the farm and managing resources throughout the crop-growing period. Soil type, climate, rainfall, irrigation availability, crop variety, previous crop, market conditions and available farm resources can all influence crop performance. Good agricultural practices provide a systematic approach for producing healthy crops while protecting soil, water, farm workers and the surrounding environment.

Good Agricultural Practices, commonly referred to as GAP, provide a framework for environmentally responsible, economically viable and socially acceptable farming. They also address food safety, worker health, efficient use of natural resources and the production of safe and good-quality agricultural products. Farmers should therefore consider the complete crop production cycle rather than treating individual activities such as fertilizer application or pesticide spraying as isolated operations.

These guidelines provide general agricultural information and are intended to support decision-making. Crop-specific recommendations can vary according to location, soil, season, variety and local agricultural recommendations. When a farmer needs an exact fertilizer quantity, pesticide dose, disease diagnosis or scheme eligibility decision, the recommendation should be verified with the relevant agricultural department, Krishi Vigyan Kendra, agricultural university, qualified agricultural expert or current official product or scheme guidance.

1. Understanding the Farm Before Cultivation

Before starting a new crop, a farmer should understand the characteristics and limitations of the field. Important factors include soil type, soil fertility, drainage, water availability, previous crop, rainfall pattern, temperature, slope of the land and the availability of farm inputs. The selection of a crop should be based on whether the crop can perform under these conditions rather than being based only on expected market price.

The previous crop can influence the nutrient status of the soil and the occurrence of pests and diseases. Continuous cultivation of the same crop or closely related crops can sometimes increase pest and disease pressure and may contribute to nutrient imbalance. Crop diversification and rotation can therefore be useful components of long-term farm management. Integrating crops with different nutrient requirements and growth patterns can improve resource use and support soil health.

Farmers should also consider the availability of labour, irrigation facilities, machinery, quality seed, fertilizers, crop protection inputs, storage facilities and access to markets. A crop that requires more water or labour than the farm can reliably provide may create avoidable production risks. Agricultural planning should therefore consider both biological and economic factors.

2. Soil Management

Soil is one of the most important resources in agriculture because it provides plants with physical support, water and nutrients. Healthy soil also contains organic matter and living organisms that contribute to nutrient cycling and soil structure. Maintaining soil health is therefore essential for long-term agricultural productivity.

Soil management should begin with an understanding of the physical and chemical properties of the field. Important soil characteristics include texture, structure, pH, electrical conductivity, organic carbon and the availability of nutrients. Soil testing can provide useful information about the nutrient status of a field and can help farmers make more informed decisions about fertilizer and soil amendments.

The Soil Health Card system provides information about important soil parameters and can provide crop-related fertilizer and soil amendment recommendations. The official Soil Health Card guidance includes nitrogen, phosphorus, potassium, sulphur, zinc, iron, copper, manganese and boron along with pH, electrical conductivity and organic carbon among the soil parameters assessed.

Farmers should avoid applying large quantities of fertilizer without understanding the nutrient requirements of the crop and the condition of the soil. Current agricultural guidance emphasizes balanced, soil-test-based fertilizer use because excessive or unbalanced fertilizer application can contribute to nutrient imbalance, soil degradation and environmental problems.

3. Soil Testing and Soil Health Assessment

Soil testing is a useful practice before establishing a crop, particularly when fertilizer recommendations need to be adjusted to field conditions. A soil sample should represent the area being cultivated and should be collected according to recommended sampling procedures. Information such as the location of the plot, crop to be grown, soil type, area, previous crop, drainage and water source can be useful in interpreting agricultural soil information.

A soil test report should not be treated simply as a list of nutrient values. Farmers should use the recommendations associated with the test results to make decisions about nutrient application and soil improvement. Soil Health Card guidance is intended to provide farmers with information about soil nutrient status along with fertilizer and amendment recommendations.

Soil health should be monitored over time rather than considered a one-time activity. Changes in organic matter, nutrient status, salinity, acidity or other soil characteristics can affect crop performance. Regular observation and appropriate soil testing can help farmers identify emerging problems and adjust management practices.

4. Land Preparation

Land preparation should create suitable conditions for seed germination, root development, water movement and crop establishment. The appropriate method depends on the crop, soil type, moisture condition and local farming system. Excessive tillage should be avoided when it does not provide a clear agricultural benefit because unnecessary soil disturbance can affect soil structure and increase erosion.

Before planting, farmers should remove or manage problematic weeds and crop residues according to the requirements of the crop. In some farming systems, healthy crop residues can be retained or incorporated to contribute organic matter and protect the soil surface. However, diseased plant material should be managed carefully so that it does not become a source of infection for the following crop.

Good land preparation should also consider drainage. Waterlogging can damage roots, reduce oxygen availability and increase the risk of some diseases. Fields that are vulnerable to water accumulation should have suitable drainage arrangements before the crop is established.

5. Selection of Quality Seed and Planting Material

The use of quality seed or healthy planting material is one of the foundations of successful crop production. Farmers should select varieties that are suitable for their local climate, soil, season, water availability and intended market. Where possible, seed should be obtained from reliable and authorized sources.

The choice of variety can influence yield, crop duration, disease resistance or tolerance, water requirement and suitability for local conditions. Farmers should not select a variety solely because it produced good results in another location because agricultural performance can vary according to climate, soil and management.

Seed should be inspected for physical quality and signs of damage, disease or insect infestation. Seed treatment may be recommended for some crops to reduce seed-borne or soil-borne problems. Any treatment should be performed using the recommended method and an approved product where applicable.

6. Seed Treatment and Crop Establishment

Seed treatment is a preventive agricultural practice that can help protect seeds and young seedlings from certain seed-borne and soil-borne problems. Depending on the crop and local recommendations, biological or chemical seed treatments may be used. Farmers should follow crop-specific recommendations rather than applying an arbitrary treatment.

Good crop establishment depends on suitable soil moisture, correct sowing depth, appropriate spacing and good-quality seed. Seeds planted too deeply may have difficulty emerging, while shallow planting may expose seeds to drying or other risks. The appropriate depth and spacing should therefore be selected according to the specific crop and variety.

Uniform crop establishment makes later irrigation, nutrient management, weed control and pest monitoring easier. Gaps in the field can reduce productivity and may also encourage uneven crop development.

7. Sowing Time and Plant Spacing

The sowing period should be selected according to crop, variety, season, rainfall pattern, temperature and irrigation availability. Sowing too early or too late can expose a crop to unsuitable weather conditions or increase pest and disease risks.

Plant spacing should provide adequate access to sunlight, air movement, water and nutrients. Very dense planting can create a humid crop canopy and may increase the risk of some diseases, while excessively wide spacing may reduce efficient use of available land. Farmers should follow crop-specific spacing recommendations from agricultural extension sources.

Proper sowing time and spacing are especially important for crops that are sensitive to temperature, rainfall or day length. Farmers should therefore consider local agricultural advisories when planning planting dates.

8. Water and Irrigation Management

Water is essential for crop growth, but both insufficient and excessive irrigation can damage crops. Irrigation should be planned according to crop stage, soil type, weather conditions, rainfall, root-zone moisture and crop water requirement.

Young plants may require careful moisture management during establishment, while crops at flowering or fruit development stages may be particularly sensitive to water stress depending on the crop. Farmers should avoid allowing the crop to experience prolonged moisture stress and should also avoid unnecessary irrigation that causes waterlogging or nutrient loss.

Efficient water management includes selecting suitable irrigation methods, maintaining irrigation equipment and preventing leakage. Drip and sprinkler systems can improve water-use efficiency in appropriate crops and situations. Water conservation measures such as mulching, rainwater harvesting, field-level moisture conservation and suitable soil management can also reduce water losses.

The suitability of an irrigation method depends on the crop, soil, terrain, water quality and available infrastructure. Farmers should therefore select irrigation systems according to local conditions rather than assuming that one method is suitable for every crop.

9. Nutrient Management

Plants require nutrients for growth, development, flowering, fruiting and seed production. The major nutrients include nitrogen, phosphorus and potassium, while secondary nutrients and micronutrients are also required in smaller quantities. Nutrient requirements vary considerably between crops and growth stages.

Nitrogen supports vegetative growth and contributes to the formation of proteins and chlorophyll. Phosphorus is involved in root development, energy transfer and reproductive development. Potassium contributes to several physiological processes and helps plants respond to environmental stresses. Secondary and micronutrients also play important roles in plant development.

Nutrient management should be based on soil-test results, crop requirement, growth stage and local recommendations. Applying more fertilizer than the crop requires does not necessarily increase yield and can create nutrient imbalance, economic loss and environmental problems. Current agricultural guidance increasingly emphasizes balanced and integrated nutrient management rather than indiscriminate fertilizer use.

10. Integrated Nutrient Management

Integrated nutrient management combines appropriate mineral fertilizers with organic sources and biological inputs where suitable. Organic sources can include well-decomposed farmyard manure, compost, crop residues and green manure. Biofertilizers may also be used when suitable for the crop and production system.

The objective is not simply to replace all chemical fertilizers with organic materials. Instead, nutrient sources should be combined in a scientifically appropriate manner to meet crop requirements while maintaining soil health. Recent agricultural guidance has emphasized the integration of chemical fertilizers with organic inputs, green manures and biofertilizers as part of sustainable soil management.

Farmers should remember that organic inputs also contain nutrients and therefore contribute to the overall nutrient budget of the farm. Their nutrient content can vary depending on source and preparation, so fertilizer decisions should consider the total nutrient contribution.

11. Organic Matter and Soil Fertility

Organic matter contributes to soil structure, water retention, nutrient cycling and biological activity. Adding suitable organic materials can improve soil quality over time, particularly when combined with appropriate crop and water management.

Farmyard manure and compost should be properly decomposed before application. Green manuring can also be useful in suitable farming systems. Legume crops may contribute to nitrogen cycling through biological nitrogen fixation and can be incorporated into crop rotations where appropriate.

Crop residues can provide organic matter and protect the soil surface when managed properly. However, residues infected with serious diseases or containing significant pest populations should be handled according to crop-specific recommendations.

12. Weed Management

Weeds compete with crops for sunlight, water, nutrients and growing space. Weed competition can be particularly harmful during the early stages of crop growth when the crop is still establishing.

Weed management should be based on the crop and the dominant weed species. Cultural methods, mechanical removal, hand weeding, mulching, crop rotation and other appropriate methods can be used. Herbicides may be used when recommended, but they should be selected and applied according to the crop, weed, product label and local agricultural guidance.

Repeated use of the same herbicide mode of action can contribute to herbicide resistance in weed populations. Integrated weed management, which combines several suitable approaches, can help reduce dependence on a single method.

13. Integrated Pest Management

Integrated Pest Management, or IPM, is an approach that combines preventive, biological, cultural, mechanical and chemical methods to manage pests while reducing unnecessary pesticide use. The first step is regular monitoring of the crop rather than automatically spraying pesticides whenever an insect is observed.

Not every insect found in a field is harmful. Some insects are beneficial predators or pollinators and can contribute to natural pest control. Farmers should therefore learn to distinguish damaging pests from beneficial organisms whenever possible.

Field sanitation, crop rotation, suitable planting material, resistant or tolerant varieties, traps, biological control and mechanical removal can all form part of an IPM programme. Chemical control may be considered when pest populations reach levels where intervention is justified and when an appropriate registered product is available.

14. Pest Monitoring

Regular field observation helps farmers identify pest problems at an early stage. Monitoring should include inspection of leaves, stems, flowers, fruits, roots and other relevant plant parts depending on the crop.

Farmers should observe the number of affected plants, the type of damage and whether the problem is increasing or remaining stable. Sticky traps, pheromone traps and other monitoring tools can be used for suitable pests.

When a pest is detected, farmers should identify the pest before selecting a control method. Spraying a pesticide without knowing the pest can waste money, harm beneficial organisms and potentially contribute to resistance.

15. Crop Disease Management

Plant diseases may be caused by fungi, bacteria, viruses, nematodes or other organisms, while some plant problems may have non-infectious causes such as nutrient deficiency, water stress, salinity, temperature stress or chemical injury. Because several causes can produce similar symptoms, visual symptoms alone may not always provide a reliable diagnosis.

Farmers should observe when the symptoms started, which plant parts are affected, whether the symptoms occur on old or young leaves, whether insects are present, whether the field has excessive moisture and whether neighbouring plants show similar symptoms.

Disease management should begin with prevention. Healthy planting material, appropriate spacing, field sanitation, crop rotation, balanced nutrition, proper irrigation and removal of infected plant material can reduce the risk of disease development.

When chemical disease management is necessary, farmers should use only currently recommended and legally approved products for the specific crop and disease. Product labels and local agricultural recommendations should be followed carefully.

16. Recognizing Nutrient Deficiency and Other Plant Problems

Yellowing, leaf spots, curling, stunted growth and premature leaf fall do not always indicate a disease. Similar symptoms can result from nutrient deficiency, excessive fertilizer, drought, waterlogging, root damage, salinity, insect feeding or viral infection.

For example, yellow leaves may occur because of nutrient deficiency, root damage or disease. A farmer should therefore consider the location of the symptom on the plant, the age of affected leaves, soil moisture, recent fertilizer application and the presence of insects before selecting a management practice.

The agricultural assistant should not treat a symptom-based question as a guaranteed diagnosis. When the available information is insufficient, the farmer should be asked for additional information such as crop name, plant age, affected plant part, symptom appearance, duration of the problem and field conditions.

17. Weather and Climate Risk Management

Weather can strongly influence agricultural production. Important weather-related risks include drought, excessive rainfall, flooding, heat stress, strong winds, cyclones and unusually low or high temperatures.

Farmers should monitor reliable local weather forecasts and agricultural advisories when planning irrigation, spraying, sowing and harvesting operations. Spraying pesticides or fertilizers immediately before heavy rainfall may result in reduced effectiveness or environmental contamination, depending on the product and application method.

In areas exposed to drought, farmers can reduce risk through water conservation, suitable crop selection, drought-tolerant varieties where available, soil-moisture conservation and efficient irrigation. In flood-prone areas, drainage planning, suitable crop selection and timely weather information can reduce production risks.

Climate-resilient agricultural practices may include crop diversification, conservation of soil organic matter, improved water management and appropriate adjustment of crop calendars.

18. Crop Diversification and Rotation

Growing different crops over time can improve the resilience of a farming system. Crop rotation can reduce the continuous build-up of certain pests and diseases and can help manage nutrient demand.

Including legume crops in a rotation can contribute to soil fertility through biological nitrogen fixation. Diversification can also reduce dependence on a single crop and may provide additional sources of income.

However, crop rotation should be designed according to local soil, climate, water and market conditions. A rotation that works well in one region may not be suitable in another.

19. Farm Drainage and Waterlogging Management

Excess water in the root zone can reduce oxygen availability and interfere with root function. Waterlogging can also create conditions favourable for some root diseases.

Fields that are naturally prone to water accumulation should be assessed before planting. Suitable drainage channels, land shaping or raised-bed systems may be considered depending on the crop and local conditions.

At the same time, drainage should be managed carefully so that valuable soil and nutrients are not unnecessarily washed away. Soil conservation and water management should therefore be considered together.

20. Farm Waste and Crop Residue Management

Agricultural activities generate crop residues, packaging materials, animal waste and other materials. Proper management of these materials is important for soil health, worker safety and environmental protection.

Healthy crop residues can sometimes be incorporated into the soil, composted or used as mulch. Diseased residues should be handled according to crop-specific recommendations because leaving infected material in the field may contribute to disease carry-over.

Plastic containers and pesticide containers should never be reused for food, water or household purposes. Waste should be disposed of according to applicable local regulations and safety guidance.

Crop-residue burning should be avoided where alternatives are available because it can cause air pollution and destroy organic matter that could otherwise contribute to soil health.

21. Farm Worker Safety

Agricultural work can involve risks from machinery, pesticides, fertilizers, animals, heat, dust and physical injuries. Farmers and farm workers should use appropriate protective equipment and follow safe operating procedures.

Pesticides should be handled only according to their approved label instructions. Workers should avoid unnecessary exposure during mixing, spraying and cleaning operations. Appropriate protective clothing should be used when required.

Pesticides should be stored in their original containers and kept away from food, drinking water, children and animals. Empty pesticide containers should not be reused for household purposes.

Workers should also be protected from heat stress by maintaining access to drinking water, taking appropriate rest periods and avoiding unnecessary exposure to extreme heat conditions.

22. Safe Use of Fertilizers and Agricultural Inputs

Fertilizers and other agricultural inputs should be stored in dry and secure locations. Different materials should be stored according to their compatibility and manufacturer instructions.

Farmers should calculate the required amount before application rather than applying fertilizer based only on visual crop appearance. Excess application can increase input costs and may damage soil, crops or the environment.

When using agricultural inputs, farmers should maintain records of the product name, quantity, date of application, crop and field. Such records can help identify the cause of problems and improve future farm planning.

23. Record Keeping

Farm records are useful for understanding farm performance over time. Farmers can maintain information about the crop variety, sowing date, seed quantity, fertilizer application, irrigation, pest and disease observations, pesticide applications, labour costs, harvest quantity and selling price.

Records can help farmers compare different production practices and identify which approaches provide better results under their specific conditions.

Digital records can also be useful. A farmer may maintain records using a mobile application, spreadsheet or notebook depending on available technology.

24. Harvest Management

Harvesting should be carried out when the crop reaches the appropriate maturity stage. Harvest timing varies considerably between crops and depends on the intended use and market requirements.

Harvesting too early can reduce quality and yield, while excessive delay can increase losses from pests, diseases, weather, shattering or deterioration. Farmers should monitor crop maturity and weather conditions when deciding the harvest date.

Harvesting equipment should be clean and properly maintained. Care should be taken to minimize physical damage to fruits, vegetables, grains and other agricultural products.

25. Post-Harvest Handling

Post-harvest management begins immediately after harvesting. Cleaning, sorting, grading, drying, packaging, transportation and storage should be carried out in a way that minimizes contamination and physical damage.

Grains and other products intended for storage should be dried to appropriate moisture levels for the particular commodity and storage system. Excess moisture can encourage fungal growth and deterioration.

Storage areas should be clean, dry, adequately ventilated where appropriate and protected from insects, rodents and other pests. Stored products should be inspected periodically so that problems can be identified early.

26. Food Safety and Quality

Agricultural products intended for human consumption should be produced and handled in a way that minimizes contamination. Farmers should use clean water where required, maintain hygienic harvesting practices and avoid contamination from chemicals, animals, waste and dirty equipment.

The correct pre-harvest interval must be observed when agricultural chemicals have been applied to crops. Farmers should follow the product label and relevant agricultural recommendations before harvesting treated crops.

Good agricultural practices connect production efficiency with food safety, worker welfare and environmental protection.

27. Sustainable Agriculture

Sustainable agriculture aims to maintain productive farming systems while protecting natural resources for the future. Soil health, water conservation, biodiversity, balanced nutrient management and efficient use of farm inputs are important components of sustainable agriculture.

Farmers can support sustainability by using soil-test-based nutrient management, conserving water, reducing unnecessary chemical applications, incorporating suitable organic matter, maintaining crop diversity and protecting beneficial organisms.

Sustainability does not mean following a single farming method everywhere. Instead, the most appropriate practices should be selected according to local environmental, economic and social conditions.

28. Natural Resource Conservation

Soil and water are limited agricultural resources and should be managed carefully. Soil erosion can remove fertile topsoil and reduce long-term productivity. Practices such as maintaining soil cover, using suitable crop rotations, reducing unnecessary soil disturbance and managing field drainage can help conserve soil.

Water conservation can include efficient irrigation, rainwater harvesting, mulching, appropriate scheduling and reduction of leakage or unnecessary water use. The correct method depends on soil, crop and local water conditions.

Natural-resource management programmes include integrated soil-water-nutrient management, watershed management, crop diversification, integrated weed management, climate-resilient agriculture and conservation agriculture among their major areas of work.

29. Managing Agricultural Risks

Every farming system has production and economic risks. Production risks may include drought, floods, pests, diseases, poor germination and extreme weather. Economic risks may include changes in input prices, market prices and demand.

Farmers can reduce risk by diversifying crops, selecting suitable varieties, maintaining emergency water arrangements where possible, monitoring weather and crop conditions, keeping farm records

and using available agricultural support services.

Crop insurance and government support programmes may also help eligible farmers manage specific risks. However, eligibility, coverage, application periods and conditions vary by scheme and should always be verified using current official information.

30. When a Farmer Should Seek Expert Advice

A farmer should seek assistance from a qualified agricultural professional when a crop problem cannot be identified confidently, when disease is spreading rapidly, when a large area of the field is affected, when plants are dying unexpectedly or when the problem involves potentially serious pesticide or chemical exposure.

Expert advice is also important when the farmer needs a precise fertilizer recommendation, pesticide dose, disease confirmation, laboratory test or interpretation of a soil test. Local agricultural departments, Krishi Vigyan Kendras, agricultural universities and qualified extension professionals can provide location-specific guidance.

An AI agricultural assistant can help farmers understand agricultural information and identify possible causes of common problems, but it should not present uncertain information as a confirmed diagnosis or replace professional field-level assessment when the situation requires it.

31. Using an Agricultural Assistant Responsibly

An agricultural assistant can help a farmer organize information about crops, soil, irrigation, nutrients, pests, diseases, weather risks and government support. The quality of its answer depends on the information provided by the farmer and the reliability of the agricultural knowledge used by the system.

When asking about a crop problem, farmers should provide as much relevant information as possible. Useful information includes the crop name, variety if known, plant age, location, growth stage, symptoms, affected plant parts, duration of the problem, irrigation conditions, recent fertilizer application, recent pesticide application and presence of insects.

For example, instead of asking only, "Why are my leaves yellow?", a more useful question would be, "My 40-day-old tomato plants have yellowing on the older leaves. The field was irrigated two days ago and I can see small insects under some leaves. What could be causing the problem?" Additional context allows an agricultural assistant to retrieve more relevant information and provide a more useful response.

32. General Decision-Making Approach for Farmers

Agricultural decisions should follow a systematic process. First, the farmer should identify the crop, field conditions and production objective. Second, the farmer should observe the current condition of the soil, crop and environment. Third, the farmer should identify possible causes of any problem rather than immediately selecting a treatment. Fourth, appropriate management options should be considered according to the crop and local recommendations. Finally, the farmer should monitor the result and adjust management when necessary.

This approach is especially important for pest and disease management. A symptom should not automatically be treated as a particular disease. The crop, symptom pattern, environmental conditions and pest presence should all be considered before deciding on a management action.

33. Summary

Successful agriculture depends on the interaction of soil, water, crops, nutrients, pests, diseases, weather and farm management. Farmers can improve the reliability of crop production by beginning with proper farm assessment, selecting suitable crops and quality planting material, maintaining soil health, managing water efficiently, applying balanced nutrients, monitoring pests and diseases, controlling weeds and preparing for weather-related risks.

Agricultural practices should be adapted to local conditions. Soil-test-based nutrient management, integrated pest management, crop diversification, water conservation, safe use of agricultural inputs, proper harvesting and post-harvest handling can contribute to productive and sustainable farming.

Farmers should use current agricultural advisories and local expert recommendations whenever a decision requires location-specific information. An agricultural assistant can support this process by providing relevant information from reliable agricultural knowledge sources, helping farmers understand possible causes of crop problems and connecting different areas of agricultural knowledge such as crop cultivation, disease management, soil management and farmer support programmes.

The objective of good agricultural practice is not simply to maximize production in a single season. It is to maintain productive soil, use water and inputs efficiently, protect farm workers and the environment, produce safe agricultural products and support the long-term sustainability and profitability of farming.

Reference and Information Sources

The agricultural principles in this document are based on publicly available guidance and information from Indian agricultural institutions and government agricultural resources. Good Agricultural Practices principles are described by the Indian Council of Agricultural Research, including environmental sustainability, economic viability, social acceptability, food safety, resource management and worker welfare.

Soil-health and soil-testing information is based on the Government of India's Soil Health Card resources, including guidance covering soil nutrient status, pH, electrical conductivity, organic carbon, macro-nutrients, secondary nutrients and micronutrients.

Guidance on balanced and integrated nutrient management is supported by current ICAR agricultural outreach and technical resources emphasizing soil-test-based fertilizer management and sustainable nutrient use.

Information concerning natural-resource management and sustainable agricultural practices is supported by ICAR resources covering soil and water conservation, integrated nutrient management, crop diversification, weed management and climate-resilient agriculture.

This document provides general agricultural guidance. Crop-specific fertilizer rates, pesticide doses, disease-control recommendations, irrigation schedules and government-scheme eligibility should be verified against current official recommendations applicable to the farmer's crop, location and season.